



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
 December 2023

ME1654-1 – INNOVATIONS AND DESIGN THINKING
 (FOR AI&DS, BT, CCE, CSE, CV, EE, RI)

Duration: 2 Hours

Max. Marks: 50

Note: Answer Three full questions choosing One full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is design thinking? Explain with an example.	07	L*1	1	2
	b) Explain 360° approach with suitable example.	08	L2	1	2
2.	a) What are the 5 design thinking principles that redefine business or business management?	07	L1	1	2
	b) How do you empathetically collect data from a person who is looking for a backpack for his/her daily use which he/she wants to carry to night trekking and wild life photography?	08	L1	1	2
Unit – II					
3.	a) With a neat sketch show agile model of project management.	07	L1	2	1
	b) With the help of a flow chart explain the process modelling in design thinking.	08	L1	2	1
4.	a) With a neat sketch show waterfall model.	07	L1	2	1
	b) What are the advantages of agile development method used in software development in IT industry?	08	L1	2	1
Unit – III					
5.	Considering the first 3 pillars of design thinking, ideate a pair of shoes which can be used for all possible purposes like daily use, jogging, (night and wetland) trekking, formal meeting, biking, family functions and also playing football.	20	L3	3	3
6.	There is a 1 acre farm land which is equivalent to 22 numbers of 30X40 Sq ft. site. This land is very close to forest and gets all the serenity and tranquility of nearby forest with some wild animals habitat like forest giant squirrel, monkeys, deer, wild boar, snakes and some varieties of native birds. The land is placed in a downhill making the rain water flow downwards and pass through the channel located at the center of the land. A corporate working engineer is looking to build a farm house on this land. He has plans of having medicinal plants in one section of the land, some fruit garden in the other section, a small lake to cultivate some fishes in the third section which can in turn serve as the source of water for the land, fourth and fifth sections for dog kennels having 3 different breeds and a cow shed holding around 5 cows, respectively. Empathize with the engineer, define his requirements and ideate the farm land layout for his ease of operation and maintenance with necessary assumptions. He is ready to invest not more than 20 lakhs on this land. Also show the respective price allotment to each section.	20	L3	3	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
